DHRUV SAINI

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EXPERIENCE

A Sustainable Future
CEO | Oct 2023 - Present
- Built and scaled the **PCM**, an AI-powered mobile app predicting paper use, to **30,000 students** nationwide in over 150 schools
- Trained a symbolic regression model with **PySR** to predict paper consumption based on school demographics
- Leading **a team of 20** to develop a nonprofit website and iterating on AI model, now helping save 1.7M sheets of paper per year
- Received **$30k** in grants supporting sustainability impact

West Virginia University: SG-REAL Lab
GIS/Software Research Intern | Jul 2025 - Present
- Created a 3D modeling and flood segmentation pipeline on any power grid to determine which nodes need most investment

Dimension
Software Engineer Intern | Jun 2025 - Sept 2025
- Implemented webhooks to allow external companies to integrate with the enterprise collaboration platform
- Led the process to allow Stripe, Vercel, Github, and many other hosted bots in channels
- Revamped and modernized the UI, completing several refactors that culminated in a **70%** reduction of load time

University of Washington
Quantum Technical Writing Intern | Aug 2024 – Sept 2024
- Worked with Prof. Anantram to develop an editing workflow for his book *Quantum Mechanics for Scientists and Engineers*
- Used Python and React to automatically compare textbook versions and allow students to easily select the best version of any paragraph
- Accelerated editing time by **15x**

PROJECTS

CAD-bench: Benchmarking Language Models on Functional CAD Generation
- Built a test suite for AI-generated 3D parts that checks whether a part works as intended, not only whether it looks correct
- Ran physics simulations of mechanisms such as gearboxes to test whether generated parts work in motion
- Accepted to the ICML AIWILD and FAGEN workshops; currently under review at NeurIPS, with code available here and paper available here

EDA Bench: An Execution-Based Benchmark for Language Model Agents on Functional PCB Construction
- Built a test suite for AI-designed circuit boards that checks electrical behavior, not only whether a design file opens
- Combined KiCad, connection and safety checks, circuit simulation, and real-world input/output requirements to test whether a circuit functions
- Accepted to the ICML FAGEN workshop; currently under review at NeurIPS, with code available here and paper available here

Residual Controllers
- Studied a small internal signal that tells an open AI model whether to begin step-by-step reasoning or give a direct answer
- On all 30 AIME 2026 problems, a one-step controller entered thinking mode on 30/30 and scored 16/30, compared with 1/30 when thinking was off; paper available here

ThinkSwitch: Iterative LoRA training and Weight Interpolation for Low-Compute Improvement on Specific-Purpose Reasoning Tasks
- Used two low-cost methods—small trainable additions and blended saved model weights—to improve Qwen3-4B on AIME with only $3 in compute
- Awarded 1st place at the Washington State Science and Engineering Fair, preprint available here

NASA Glenn High School Engineering Institute | Selected Participant
- Designed and tested jet-noise, spacecraft-power, and lunar-rover prototypes

AWARDS AND HONORS

National Science Bowl
Team Captain | 2021-Present
- Founded Bellevue High School's first science bowl team, winning 1st place at the nationwide virtual competition
- Won a fully-funded trip to Washington D.C. for the National Finals

Olympiad Awards
Various | 2024-Present
- USACO Platinum Rank
- USAAIO National Finalist, Bronze Medal

ARC - The American Rocketry Challenge
Team Captain | 2025
- Captained a team of five to build a model rocket following specific criteria, qualified for National Finals twice

EDUCATION

Bellevue High School
- 36 ACT, 100+ hours of community service, DECA ICDC finalist
- President of Programming Club, coach at Tyee MS and Odle MS programming clubs